

Phase 2 Design Document

Purpose

Implement the Exercise, Health, Mental Wellness, and Nutrition modules using a compact layered architecture that is easy to teach and extend.

Architecture

Streamlit page classes render the interface. Dataclasses represent records. Repository classes own all SQL. AnalyticsService performs reusable calculations. DatabaseManager controls SQLite connections and transactions.

Database

Each domain table references clients.id. Records use soft deletion through is_active. Date and client indexes support history and trend queries.

Permissions

Administrators create, view, and deactivate records. Clients create and view records but cannot change historical data.

Phase boundary

Cross-domain correlations are Phase 3. Excel import, PDF and Excel exports, and scheduled reports are Phase 4.